COVER — Project index and design basis

NodeSnap · NodeSnap Villa — NYC demonstration · Revision 1 (41ba6b04-9f1a-4339-9182-c295fd4a5639)
Professional review draft — Demonstration design, assumptions require professional verification

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Product promise: homeowner intent -> coordinated editable design -> traceable calculations -> professional review package.
Site mode: synthetic_demo — Demo Villa NYC-01; real address: none; BBL: none (synthetic demonstration site, 30 x 45 m context)
Jurisdiction: New York City · profile 2026-10-26-baseline
Construction codes: 2022 NYC Construction Codes (Building, Mechanical, Plumbing, Fuel Gas), with amendments
Structural loading basis: ASCE 7-16 subject to NYC Building Code Chapter 16 modifications (not ASCE 7-22)
Energy: 2025 NYCECC for complete applications filed on or after 30 March 2026 (residential pathway to be confirmed per project)
Electrical: 2025 NYC Electrical Code (NYC amendments to 2020 NEC), general changeover 21 December 2025
Authority process: Not submitted. No job number, signature, seal, or approval is generated by NodeSnap.
Sheet index: COVER (this sheet) · C-101 site context · A-101..A-104 floor plans · CALC-101 pool + area schedule · S-101 structure · F-101 foundations · MPE-101 services (M/P/E/EN) · REG-101 registers
Approval fields: DOB job number [intentionally blank] · Professional signature/seal [intentionally blank — never simulated]
Status vocabulary: method_state / execution_status / check_outcome / freshness / review_status are independent dimensions; a generated sheet never upgrades a check.

Discipline result summary for this revision:
- pool: method=pool_quantities state=implemented execution=completed check=not_evaluated review=unreviewed
- architecture: method=area_schedule state=implemented execution=completed check=not_evaluated review=unreviewed
- structure: method=structural_frame_analysis state=implemented execution=completed check=not_evaluated review=unreviewed
- foundations: method=foundation_bearing_review state=implemented execution=completed check=not_evaluated review=unreviewed
- hvac/plumbing/electrical/energy/drainage: method=services_inventories state=implemented execution=completed check=not_evaluated review=unreviewed

C-101 — Site context (synthetic demonstration)

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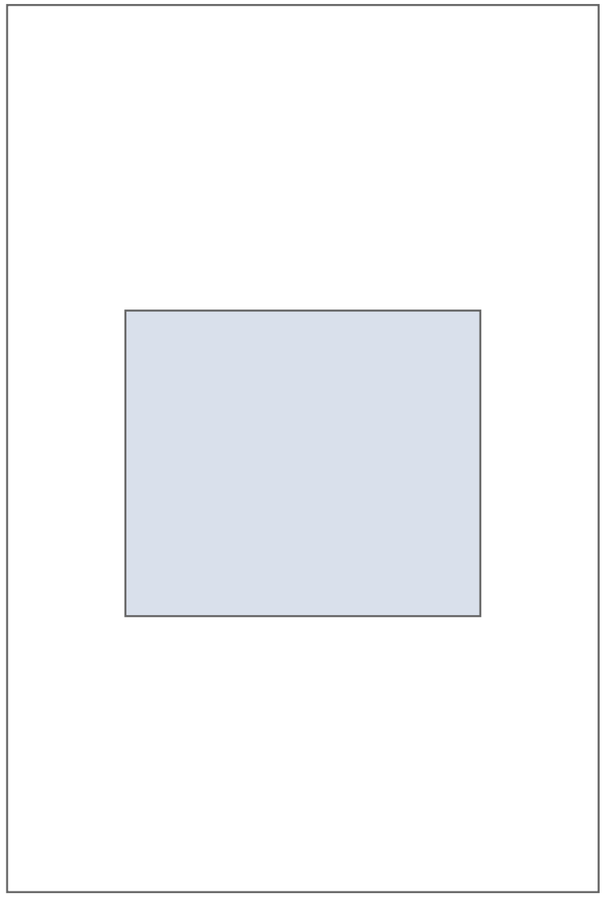

Synthetic 30 x 45 m context (scenario parameter, not a surveyed lot). Building envelope 18.0 x 15.5 m.

No real address, no BBL, no surveyed boundary, no zoning determination is claimed for this demonstration.

Real-parcel mode (spec D2/D3 adapters: GeoSearch, PLUTO/MapPLUTO, zoning layers, USGS seismic, FEMA/NYC flood screening) remains to be integrated; until then site-specific checks are not evaluated.

Unknowns preserved as unknowns: soil, groundwater, easements, utilities, flood determination, historic/special-district status.

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Level: basement Revision: 1 (41ba6b04-9f1a-4339-9182-c295fd4a5639)

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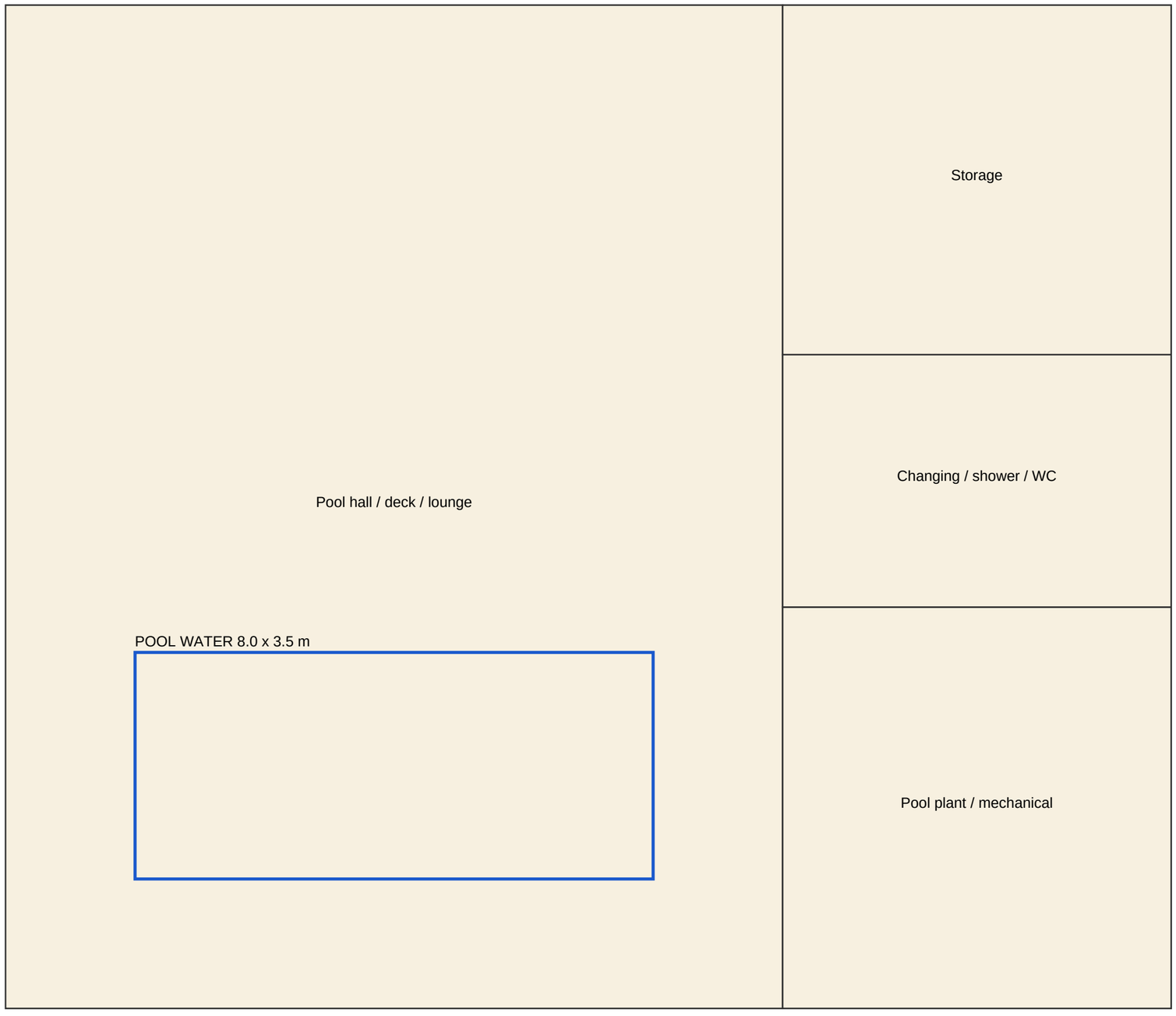

Scale: drawing fitted to page; envelope 18.0 x 15.5 m. Units: m (SI canonical). North: +Y.

NodeSnap — NodeSnap Villa — NYC demonstration

Level: ground Revision: 1 (41ba6b04-9f1a-4339-9182-c295fd4a5639)

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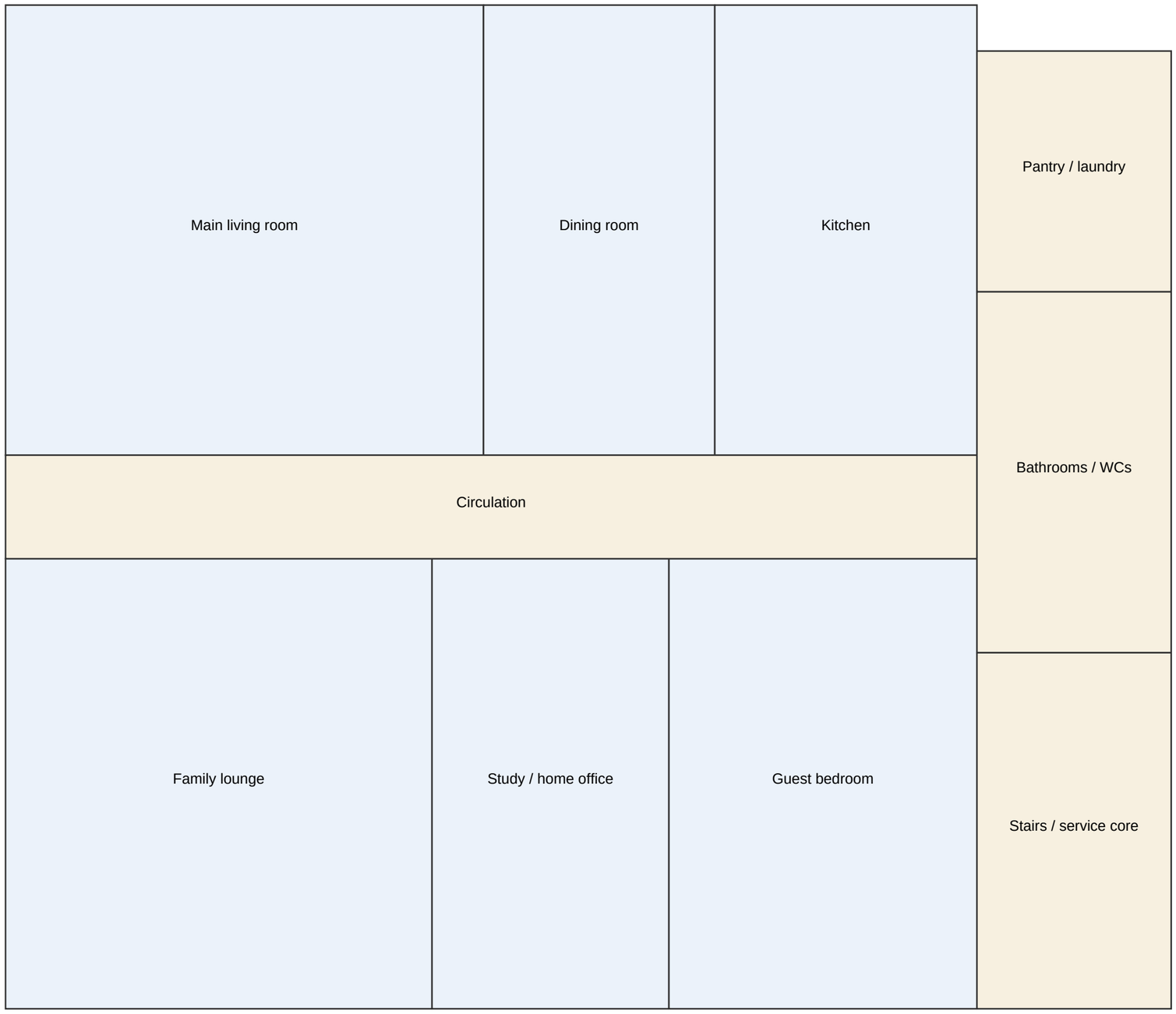
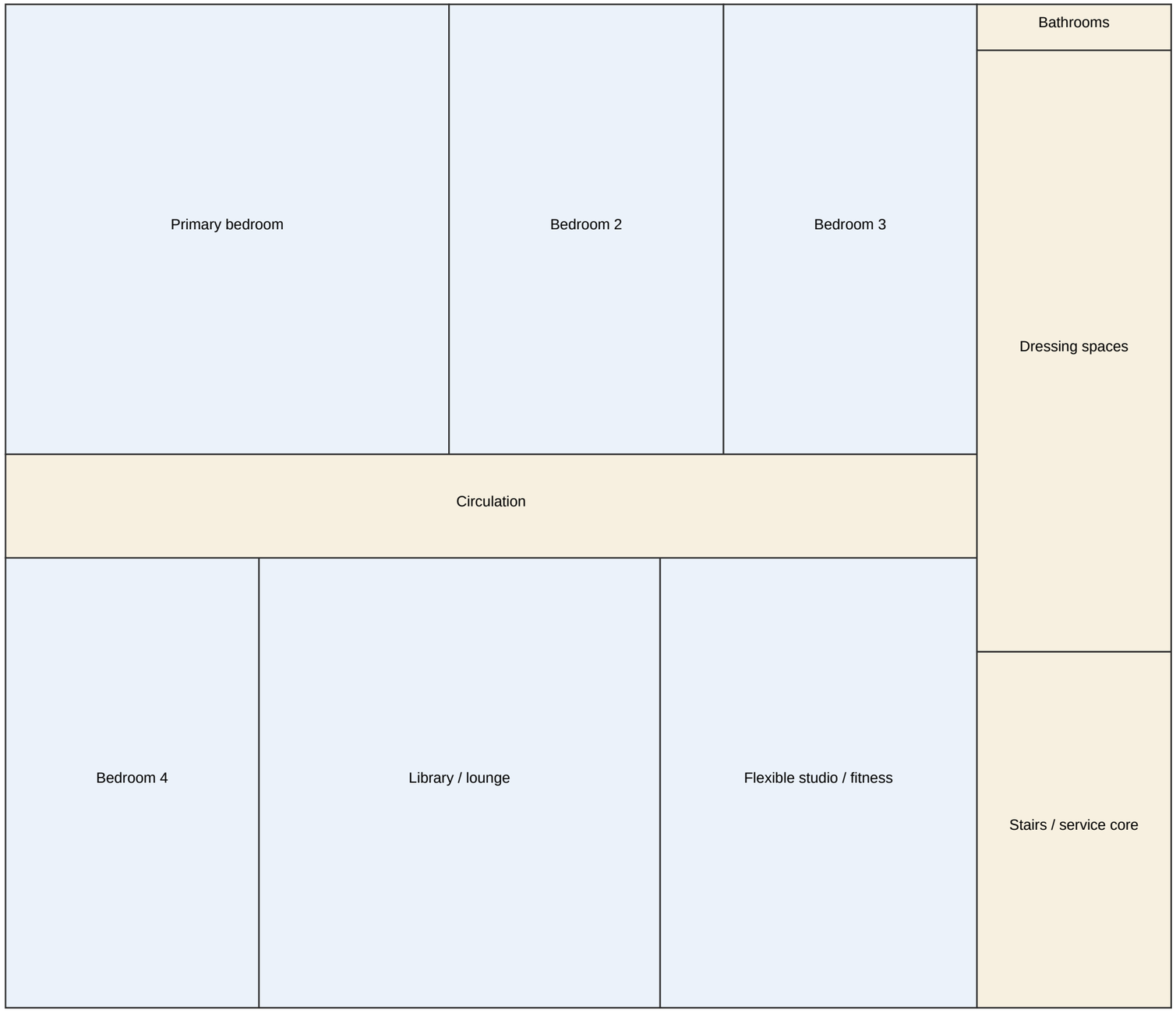


Scale: drawing fitted to page; envelope 18.0 x 15.5 m. Units: m (SI canonical). North: +Y.

NodeSnap — NodeSnap Villa — NYC demonstration

Level: upper Revision: 1 (41ba6b04-9f1a-4339-9182-c295fd4a5639)

Professional review draft — Demonstration design, assumptions require professional verification



Scale: drawing fitted to page; envelope 18.0 x 15.5 m. Units: m (SI canonical). North: +Y.

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Level: roof Revision: 1 (41ba6b04-9f1a-4339-9182-c295fd4a5639)

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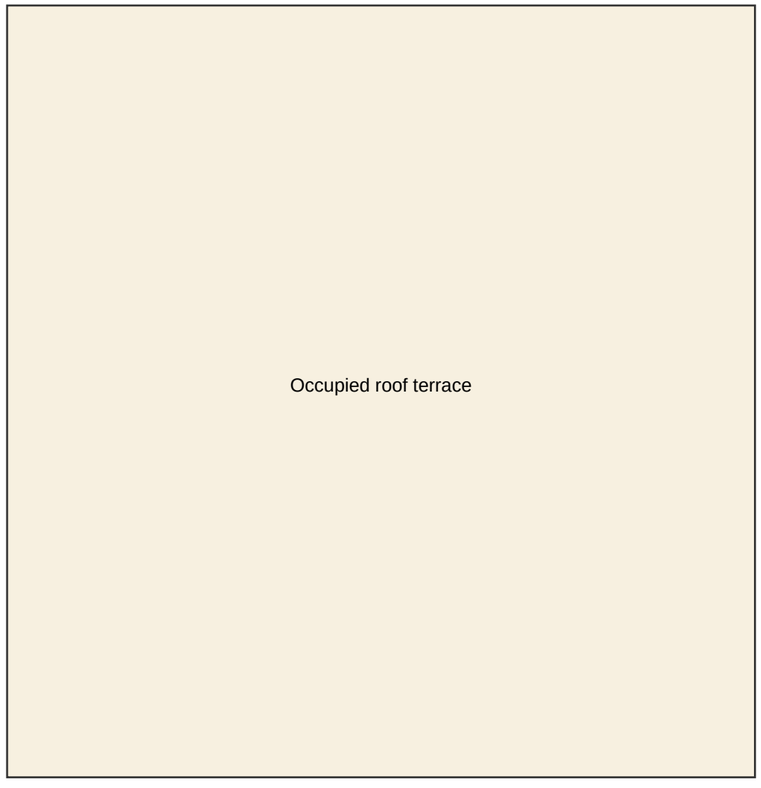

Scale: drawing fitted to page; envelope 18.0 x 15.5 m. Units: m (SI canonical). North: +Y.

Calculation & schedule — pool quantities and areas

Revision 1 (41ba6b04-9f1a-4339-9182-c295fd4a5639) content_hash=d1e7e5be19d0ae13…

Pool water: 8.0 x 3.5 x 1.4 m (flat-bottom declared basin)

Surface area = 28.0000 m2

Volume = 39.2000 m3

Mass = 39200.00 kg (rho = 1000.0 kg/m3)

Weight = 384420.68 N (g = 9.80665 m/s2)

Max hydrostatic pressure = 13729.31 Pa

Method: pool_quantities (implemented; physics only — not pool-shell design). Input basis: demonstration_assumption.

Area schedule (actual from canonical geometry vs program target)

Ground floor	Main living room	actual	51.30 m2	target	57.60	delta	-6.30
Ground floor	Dining room	actual	24.82 m2	target	27.87	delta	-3.05
Ground floor	Kitchen	actual	28.13 m2	target	31.59	delta	-3.46
Ground floor	Family lounge	actual	45.77 m2	target	33.45	delta	+12.32
Ground floor	Study / home office	actual	25.43 m2	target	18.58	delta	+6.85
Ground floor	Guest bedroom	actual	33.05 m2	target	24.15	delta	+8.90
Ground floor	Circulation	actual	24.00 m2	target	—	delta	—
Ground floor	Stairs / service core	actual	16.50 m2	target	—	delta	—
Ground floor	Bathrooms / WCs	actual	16.72 m2	target	16.72	delta	+0.00
Ground floor	Pantry / laundry	actual	11.15 m2	target	11.15	delta	-0.00
Upper floor	Primary bedroom	actual	47.59 m2	target	39.02	delta	+8.57
Upper floor	Bedroom 2	actual	29.46 m2	target	24.15	delta	+5.31
Upper floor	Bedroom 3	actual	27.20 m2	target	22.30	delta	+4.90
Upper floor	Bedroom 4	actual	27.20 m2	target	22.30	delta	+4.90
Upper floor	Library / lounge	actual	43.06 m2	target	35.30	delta	+7.76
Upper floor	Flexible studio / fitness	actual	33.99 m2	target	27.87	delta	+6.12
Upper floor	Circulation	actual	24.00 m2	target	—	delta	—
Upper floor	Stairs / service core	actual	16.50 m2	target	—	delta	—
Upper floor	Dressing spaces	actual	27.87 m2	target	27.87	delta	+0.00
Upper floor	Bathrooms	actual	2.13 m2	target	33.45	delta	-31.32
Basement	Pool hall / deck / lounge	actual	186.00 m2	target	139.35	delta	+46.65
Basement	Pool plant / mechanical	actual	37.20 m2	target	37.16	delta	+0.04
Basement	Changing / shower / WC	actual	23.40 m2	target	23.23	delta	+0.17
Basement	Storage	actual	32.40 m2	target	23.23	delta	+9.17
Roof / terrace	Occupied roof terrace	actual	83.61 m2	target	83.61	delta	-0.00

S-101 — Structure — baseline gravity frame

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Idealization: 3-bay x 3-bay steel gravity frame, fixed bases at z=0, rigid joints, slab load transferred to X-beams by tributary width (Y-beams carry no area load)

Total gravity load applied: 3,069,000 N; sum of base reactions: 3,069,000 N; equilibrium residual: 1.52e-16

Maximum beam moment (magnitude): 102,288 N-m; maximum beam deflection: 45.8 mm

Loads and section properties are explicit demonstration assumptions (see REG-101). Self-weight excluded in this scenario and not double counted.

Checks NOT evaluated: steel/timber member design, connections, diaphragm, lateral/seismic system, serviceability limits — sourced methods still gated. This sheet reports elastic responses only.

Base reactions by column (N):

(0.0,0.0): 81,443
(0.0,5.1667): 162,953
(0.0,10.3333): 162,953
(0.0,15.5): 81,443
(6.0,0.0): 174,412
(6.0,5.1667): 348,443
(6.0,10.3333): 348,443
(6.0,15.5): 174,412
(12.0,0.0): 174,412
(12.0,5.1667): 348,443
(12.0,10.3333): 348,443
(12.0,15.5): 174,412
(18.0,0.0): 81,443
(18.0,5.1667): 162,953
(18.0,10.3333): 162,953
(18.0,15.5): 81,443

F-101 — Foundations — bearing scenario

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Pressure assumption: uniform average pressure = total base reaction / footprint area

Average pressure: 11,000.0 Pa vs assumed allowable 100,000 Pa -> utilization 0.110

Missing input: project geotechnical basis (MISSING_INPUT). NYC BC 1803.2 requires a geotechnical investigation for a real structure; this scenario cannot satisfy it.

Not evaluated: settlement, uplift, sliding, overturning, punching, reinforcement, excavation

MPE-101 — Services — mechanical / plumbing / electrical / energy / drainage

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Electrical: connected load 24,254 W; current at declared 240.0 V = 101.1 A. Connected load only; demand factors, ampacity, protection and service sizing are research_required.

Plumbing: 11 fixtures; summed fixture flows 74 L/min. Arithmetic sum of declared fixture flows; fixture-unit/simultaneity sizing is research_required.

Energy/HVAC envelope conduction at dT=22.0 K: total 6,383 W (assumed U-values; component only).

- Above-grade walls: A=455.6 m2, U=0.3 W/m2K, Q=3,007 W
- Roof: A=279.0 m2, U=0.2 W/m2K, Q=1,228 W
- Ground/basement envelope (simplified): A=279.0 m2, U=0.35 W/m2K, Q=2,148 W

Drainage: roof runoff scenario 7.36 L/s (C=0.95, rainfall 100.0 mm/h, both assumed). Scenario quantity only; gutter/downpipe/detention/connection sizing is research_required.

Pool environment/evaporation, ventilation rates, equipment selection, demand factors, ampacity, panel schedules, NYCECC pathway and hydraulic sizing: research_required — no values invented for this sheet.

REG-101 — Registers — assumptions, methods, open issues

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Assumptions (all demonstration_assumption unless stated):

- floor_to_floor_height_m = 3.4 — Demonstration concept height; legal classification not determined.
- basement_elevation_m = -3.2 — Local reference elevation for concept generation only.
- pool.water_dimensions_m = [8.0, 3.5, 1.4] — Spec A3.2 flat-bottom conceptual pool; editable.
- site.context_m = [30.0, 45.0] — Synthetic Demo Villa NYC-01 context; not a surveyed lot.

Methods and coverage:

- pool_quantities: state=implemented, execution=completed, check=not_evaluated
- area_schedule: state=implemented, execution=completed, check=not_evaluated
- structural_frame_analysis: state=implemented, execution=completed, check=not_evaluated
  OPEN [UNVERIFIED_SOURCE] member_design_checks: Elastic responses only; steel/timber member, connection and lateral checks require sourced methods.
- foundation_bearing_review: state=implemented, execution=completed, check=not_evaluated
  OPEN [MISSING_INPUT] geotechnical_design_basis: No project geotechnical basis; scenario uses an assumed 100 kPa allowable bearing.
- services_inventories: state=implemented, execution=completed, check=not_evaluated
  OPEN [UNVERIFIED_SOURCE] code_sizing_methods: Inventories and component quantities only; code sizing/ventilation/energy-pathway/hydraulic methods remain research_required.

Professional review: unreviewed. Authority process: not_submitted. Signature/seal/job-number fields are intentionally blank.